# Statistical Power- Lab 5.2

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**Topics**

- Controlling Type I error
- Controlling Type II error
- Power and sample size calculations

Most studies are done to establish evidence in favor of an alternative hypothesis. The **power** of a statistical test is the probability that the test will reject the null hypothesis when the alternative hypothesis is true. Power depends on the hypothesized difference between two population means ($|\mu_1 - \mu_2|$), the population standard deviation of each group ($\sigma_1$, $\sigma_2$), and the sample size of each group ($n_1$, $n_2$). Usually, a study team can only control sample size.

The power of a test can be expressed as $P(\text{reject } H_0 | H_A \text{ is true}) = 1 - \beta$, where $\beta$ is the probability of making a Type II error (failing to reject $H_0$ when $H_A$ is true). Recall that the significance level of a test, $\alpha$, is the probability of making a Type I error (rejecting $H_0$ when $H_0$ is true).

| | **Result of test** | |
|---|---|---|
| **State of nature** | **Reject $H_0$** | **Fail to reject $H_0$** |
| $H_0$ is true | Type I error, probability $= \alpha$ (false positive) | No error, probability $= 1 - \alpha$ (true negative) |
| $H_A$ is true | No error, probability $= 1 - \beta$ (true positive) | Type II error, probability $= \beta$ (false negative) |

This lab uses simulation to explore how Type I and Type II error are controlled, and examines factors influencing the power of a statistical test. The last section introduces the formulas for power and sample size calculations in the two-group setting.

## Introduction

Suppose a pharmaceutical company has developed a new drug for lowering blood pressure and is planning a clinical trial to test the drug's effectiveness. Participants are randomized to one of two treatments, either a currently accepted medication or the new drug. At the end of the study, a hypothesis test will be conducted to assess whether there is evidence that the new drug performs better than the standard medication.

The following sections examine simulations performed under two scenarios: 1) the null hypothesis is true, and there is no difference in population mean blood pressure between the two groups, or 2) the alternative hypothesis is true, and there is a difference in population mean blood pressure between the two groups. For clinical trial data, it is standard practice to test the two-sided alternative.

Note that for this setting, there is no actual population of individuals taking the new drug (since the drug is not yet available on the market). Regardless, the observations on the participants assigned to take the new drug are treated as if they are a random sample from a hypothetical population.

### Controlling Type I error

Suppose that the null hypothesis $H_0 : \mu_{treatment} = \mu_{control}$ is true. Let the mean systolic blood pressure in both groups be 140 mm Hg, with standard deviation of 10 mm Hg; assume that blood pressures are normally distributed.

1. Run the following code to simulate blood pressure values for 50 individuals in the control group and 50 individuals in the treatment group, stored in the vectors `control` and `treatment`.

The `rnorm()` function draws `n` random numbers from a normal distribution with a given mean and standard deviation.

```
#set the parameters
control.mean = 140
treatment.mean = 140
control.sigma = treatment.sigma = 10
control.n = treatment.n = 50

alpha = 0.05

#set seed
set.seed(2018)

#simulate data
control = rnorm(n = control.n, mean = control.mean, sd = control.sigma)
treatment = rnorm(n = treatment.n, mean = treatment.mean, sd = treatment.sigma)

#conduct the test
```

a) Calculate $\overline{x}$ and $s$ for `control` and `treatment` to confirm that the simulated values seem plausible given the specified parameters for $\mu$ and $\sigma$. Would you expect $\overline{x}$ and $s$ to be exactly the same as the parameter values? Why or why not?

b) Using `t.test()`, conduct a two-sided test of the null hypothesis from the simulated data. Summarize your conclusions.

2. Run the following code to repeat the simulation 1,000 times. With each iteration, the code draws a new set of control and treatment values, conducts the two-sample $t$-test, and records the $p$-value. The logical vector `reject` records whether the $p$-value for a particular iteration was significant at $\alpha$ (i.e., less than or equal to $\alpha$).

```
#set parameters
control.mean = 140
```

```r
treatment.mean = 140
control.sigma = treatment.sigma = 10
control.n = treatment.n = 50

alpha = 0.05
replicates = 1000

#set seed
set.seed(2018)

#create empty list
p.values = vector("numeric", replicates)

#run simulations
for (k in 1:replicates){
  control = rnorm(n = control.n, mean = control.mean, sd = control.sigma)
  treatment = rnorm(n = treatment.n, mean = treatment.mean, sd = treatment.sigma)

  p.values[k] = t.test(control, treatment, alternative = "two.sided", mu = 0,
       conf.level = 1 - alpha)$p.val
}

#view results
reject = (p.values <= alpha)
table(reject)
```

a) With 50 individuals in each group, what percentage of tests result in the (incorrect) conclusion that the two population means are different?

b) Does Type I error rate change with sample size? Modify the simulation code to assess sample sizes of 100, 1,000, and 10,000.

### Controlling Type II error

Now, suppose that the alternative hypothesis $H_A : \mu_{treatment} \neq \mu_{control}$ is true. Let the mean systolic blood pressure be 140 mm Hg in the control group and 138 mm Hg in the treatment group.

3. Run the code shown in the template to simulate blood pressure values for 25 individuals in the control group and 25 individuals in the treatment group.

a) Conduct a two-sided test of the null hypothesis from the simulated data. What is the conclusion of the test?

b) Is the conclusion from part a) correct?

4. *Power and Sample Size.* Run the code chunk shown in the template to repeat the simulation 1,000 times.

a) With 25 individuals in each group, how many tests result in the incorrect conclusion that the two population means are not different?

b) Estimate the power of the two-sample test when each group has $n = 25$, $\sigma = 10$, and $\mu_{treatment} - \mu_{control} = -2$ mm Hg. Recall that power refers to the probability of rejecting $H_0$ when $H_A$ is true.

c) How does power change with increasing sample size? Estimate the power of the test as $n$ changes to 50, 100, 200, and 300 (leaving all other parameters the same).

d) Run the code chunk shown in the template to create a plot of power against sample size. The code uses the command `power.t.test()` to calculate power; this command will be discussed later in the lab. Does power seem linear in relation to sample size?

5. *Power and Standard Deviation.* For simplicity, this simulation assumes that the standard deviation of the treatment and control groups are equal.

a) Would you expect the probability of rejecting $H_0$ when $H_A$ is true to increase or decrease if there is more variation in the observations? (*Hint*: Consider the formula for the test statistic in the independent two-group setting.)

b) How does power change with increased standard deviation? Estimate the power of the test as the standard deviation within each group changes to 10, 15, and 20 (leaving all other parameters the same).

6. *Power and Effect Size.* The population effect size refers to the difference between the population means, $\mu_{treatment} - \mu_{control}$. In the simulations so far, $\mu_{treatment} - \mu_{control} = 138 - 140 = -2$ mm Hg.

In a realistic setting, the effect size is chosen to be the incremental value of the intervention that would justify changing current clinical recommendations from an existing intervention to a new one. The simulations so far mimic a setting in which researchers decide they are interested in detecting an effect on blood pressure that is 2 mm Hg or greater, when comparing the new drug to the old drug.

a) If the true difference in the group means is relatively large (e.g., 5 mm Hg), as opposed to relatively small (e.g., 1 mm Hg), would you expect the probability of rejecting $H_0$ when $H_A$ to be relatively large or relatively small?

b) How does power change with effect size? Estimate the power of the test as effect size increases; change `treatment.mean` from 138 to 137, 136, and 135 (leaving all other parameters the same).

**Power and sample size calculations**

In practice, power and sample size can be calculated directly from formulas. While hand calculations can provide quick estimates in the early stages of planning a study, statistical software should be the method of choice for a formal analysis.

*Calculating sample size for comparing two means*

$$n = \frac{(\sigma_1^2 + \sigma_2^2)(z_{1-\alpha/2} + z_{1-\beta})^2}{\Delta^2},$$

where $n$ is the number of participants in each group, $\Delta = \mu_1 - \mu_2$ is the effect size, $z_{1-\alpha/2}$ is the point on a standard normal distribution with area $1 - \alpha/2$ to its left, and $z_{1-\beta}$ is the point on a standard normal distribution with area $1-\beta$ to its left. The null hypothesis of the test is $H_0 : \Delta = 0$, tested against the alternative hypothesis $H_A : \Delta \neq 0$.

*Calculating power for comparing two means*

$$1 - \beta = P\left(Z < -z_{1-\alpha/2} + \frac{\Delta}{\sqrt{\sigma_1^2/n + \sigma_2^2/n}}\right),$$

where $Z$ is a standard normal random variable and the study has $n$ participants in each group.

The `power.t.test()` function in R can both compute the power of a one- or two-sample $t$-test (given the appropriate parameters) and determine necessary parameters (e.g., sample size) to obtain a target power. Specific instructions for using `power.t.test()` can be found in the template, the lab notes for this unit, and in the R help file.

7. A pharmaceutical company has developed a new drug to lower blood pressure and is planning a clinical trial to test its effectiveness. Individuals whose systolic blood pressures are between 140 and 180 mm Hg will be recruited for the study. Based on previous published studies, it is estimated that the patients' blood pressures will be approximately symmetrically distributed, with standard deviation of about 12 mm Hg.

The participants will be randomly assigned to the new drug or a standard drug and at the end of the study their systolic blood pressures will be measured. The company expects to receive FDA approval for the drug if there is evidence at $\alpha = 0.05$ that in the general population of people with blood pressure in the same range, the drug lowers blood pressure, on average, by at least 3 mm Hg more than the standard drug.

a) How large should the study be if the company wants the power of the study to be 80%?

b) What would the power of the study be if 200 individuals were recruited for each group?

c) Does $\alpha$ influence power? What would the power of the test be in part b) if $\alpha$ increased to 0.10? What if $\alpha$ decreased to 0.01?